Entry-Timing Confirmation via Maximum Adverse Excursion: An Empirical Test on a Systematic Index Futures Strategy

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Conflict of interest disclosure: The author operates MTS Institute and a subscription based market research publication, The Regime Report. The systematic signal tested in this paper is a separate, proprietary strategy and is not part of the publicly marketed research framework used in those products. This paper is offered as independent, self funded research and does not constitute investment advice or a recommendation to trade any instrument.

Abstract

Systematic trading research overwhelmingly concentrates on signal generation, the rules that determine direction, while treating the moment of execution as a fixed consequence of the signal rather than a decision variable in its own right. This paper isolates entry timing as a distinct layer and tests one specific, easily implementable variant: delaying entry until price has moved a threshold percentage against the signal's direction (a pullback for long entries, a rally for short entries), using each trade's recorded Maximum Adverse Excursion (MAE, Sweeney, 1996) to determine, ex post, whether such a delayed limit order would have filled. Applied to a proprietary systematic trend-continuation signal traded on NSE Nifty index futures over a fifteen-year sample (2010 to 2026, N = 3,908 trades), a modest pullback-confirmation requirement (approximately 0.25% of the underlying) improves the Sharpe ratio from 1.38 to 3.33 and the Sortino ratio from 2.59 to 10.11, net of realistic transaction

costs, while retaining 63% of the original trade population. The improvement is robust across a threshold band rather than concentrated at a single point, and Monte Carlo resampling confirms the result is not an artefact of favorable trade ordering. We discuss the likely mechanism (adverse-selection filtering of low-conviction signals), the opportunity cost (profitable trades foregone when the required pullback never materializes), and the boundary conditions under which this technique should be expected to generalize.

Keywords: entry timing, maximum adverse excursion, systematic trading, execution, limit orders, trend following

AI Disclosure: The author used Claude (Anthropic), an AI assistant, for data analysis, statistical computation, and drafting assistance in the preparation of this paper. All research design, interpretation, and conclusions are the author’s own.

1. Introduction

The academic and practitioner literature on systematic trading is asymmetric in an easily overlooked way. A large body of work addresses signal construction (which conditions should trigger a trade) and a separate, largely disconnected body of work addresses optimal execution of a decision already made, primarily in the market microstructure literature on order placement and slicing (Almgren & Chriss, 2000; Guo, de Larrard, & Ruan, 2017). Comparatively little attention sits between these two layers: given that a signal has fired, is the instant of firing actually the optimal moment to act on it, or is there a cheap, testable, rule-based way to wait for confirmation without forfeiting the edge the signal was designed to capture?

This question is distinct from execution-cost minimization in the microstructure sense. It is not about how to fill a given order at the lowest impact cost; it is about whether the order should be placed at all at the moment the signal fires, versus at a specific,

worse price that the signal must first reach. This is closer in spirit to discretionary trading heuristics (“don’t chase, let it come to you”) than to formal execution theory, and to our knowledge it has not been subjected to a rigorous, cost-adjusted, out-of-sample-style statistical test using a real multi-year trade population and a well-established excursion metric.

We use Maximum Adverse Excursion (MAE), introduced by Sweeney (1996, 1997) as a tool for empirically calibrating stop-loss placement, for a different purpose: to determine, for a completed historical trade, whether a hypothetical resting limit order at a specified adverse offset from the original signal price would have been filled before the trade’s actual exit. This allows a clean, non-parametric simulation of a delayed-entry rule without requiring tick-level order book reconstruction.

1.1 Contribution

This paper makes three contributions. First, it formalizes a simple, MAE-based method for testing delayed, confirmation-based entry rules on any existing trade log, requiring only entry price, exit price, and per-trade MAE, data that most trade-management platforms already record. Second, it provides a full empirical test on a real, multi-year systematic strategy, including realistic transaction costs and Monte Carlo robustness checks, rather than a stylized illustration. Third, it documents that the effect, where present, is not knife-edge; the improvement holds across a range of thresholds, which is evidence against the result being a curve-fitting artefact.

1.2 A Note on Disclosure

The underlying trading signal to which this entry-timing overlay is applied is proprietary and its specific rules are withheld, consistent with standard practice for practitioner-oriented empirical work of this kind. What is fully disclosed, and what this paper’s contribution rests on, is the entry-timing

methodology itself (Section 3.3), which is signal-agnostic and can be applied to any systematic strategy with a recorded MAE per trade, together with the complete statistical results of applying it. A reader cannot reconstruct the underlying signal from the material presented here; they can, however, fully reconstruct and test the entry-timing technique on their own strategy.

2. Related Work

Sweeney (1996, 1997) introduced Maximum Adverse Excursion as a diagnostic for calibrating stop-loss levels: by plotting MAE against final trade outcome across a large trade sample, a trader can identify the adverse-excursion level beyond which trades rarely recover, informing where a protective stop should sit. Subsequent practitioner literature has applied MAE (and its counterpart, Maximum Favorable Excursion, MFE) primarily to exit-side decisions, stop placement and profit-target calibration, with comparatively little published work using MAE to inform entry-side decisions.

The market microstructure literature offers a related but distinct line of work on optimal order placement in a limit order book (Guo, de Larrard, & Ruan, 2017; Almgren & Chriss, 2000), which typically takes the decision to trade as given and optimizes the mechanics of execution, order size, timing within a fixed window, and limit versus market order mix, to minimize cost and market impact. This literature generally does not address whether the initiating signal itself should be delayed, since the models typically begin from a mandate to complete a given trade within a horizon (e.g., liquidating a position), rather than from a discretionary systematic signal that could simply be skipped if not confirmed.

Discretionary trading heuristics around waiting for a pullback or retracement before entering a trend are widely discussed in practitioner and educational trading literature (e.g., retracement-based entry frameworks common in technical analysis), but this literature is generally not statistically rigorous: claims are illustrated with selected chart examples rather than tested across a full trade population with cost adjustment and robustness analysis. This paper aims to bring the same empirical standard applied to signal research to this specific, execution-adjacent question.

3. Data and Methodology

3.1 Data

The sample consists of 3,908 closed round-trip positions generated by a proprietary systematic trend-continuation signal traded on NSE Nifty 50 index futures, hourly bar resolution, from 13 April 2010 to 30 January 2026 (approximately 15.8 years). Each record includes the signal direction (long or short), entry price and timestamp, exit price and timestamp, and the trade's Maximum Adverse Excursion, defined as the worst price the position touched (in the position's unrealized loss direction) at any point between entry and exit. All returns are computed as a percentage of entry price rather than in index points, ensuring comparability across the roughly fivefold range the underlying index traversed over the sample period.

3.2 Base Strategy

The base signal's specific entry and exit rules are proprietary and are not disclosed in this paper (Section 1.2). For the purpose of this study it should be treated as a black box that produces a sequence of directional trade decisions, each with a known entry price, exit price, and intra-trade MAE; the overlay methodology tested here does not depend on, and is not specific

to, any detail of how that signal is generated. The base signal's own unfiltered, cost-adjusted performance (Sharpe ratio 1.38, annualized) is reported in Section 4.1 purely as the baseline against which the entry-timing overlay is evaluated.

3.3 Pullback-Confirmation Overlay

For a chosen threshold θ (expressed as a percentage of the entry price), a trade is redefined as follows:

- If the trade's MAE was at least as deep as θ (i.e., price moved against the signal by θ or more at some point before the original exit), the trade is retained. Its entry price is reset to the level θ away from the original signal price in the adverse direction, and its return is recalculated from that revised entry to the original, unchanged exit price.
- If the trade's MAE never reached θ, the trade is excluded from the filtered sample entirely, on the interpretation that a resting limit order placed at the θ offset would never have been filled; price moved directly in the signal's favor without offering the required confirmation.

This is directly and realistically implementable as a live trading rule: upon signal generation, submit a limit order at the entry price adjusted by θ against the signal direction, with a time-in-force matching the strategy's typical holding horizon, and cancel if unfilled. The method assumes that a resting limit order at the θ level would fill without additional slippage once price trades through it, and that price moves continuously through the θ level en route to the recorded MAE, both standard, defensible assumptions at hourly bar resolution, though not certainties.

3.4 Transaction Costs

All performance figures reported net of cost apply a round-trip transaction cost of 6.02 basis points per trade, comprising India-specific statutory charges (Securities Transaction Tax, exchange transaction charges, stamp duty, applicable at the time of writing) and an assumed slippage allowance of two index points, consistent with typical execution quality for hourly-bar-frequency systematic trading on this instrument. This cost is subtracted from every trade in both the baseline and filtered samples, ensuring the comparison in Section 4 is not an artefact of ignoring execution frictions.

3.5 Performance Metrics and Robustness Testing

Standard performance metrics are reported: compound annual growth rate (CAGR), annualized Sharpe ratio (monthly returns, risk-free rate = 0), annualized Sortino ratio (downside deviation only), Calmar ratio (CAGR divided by maximum drawdown), maximum drawdown, win rate, and profit factor. Robustness is assessed via Monte Carlo simulation: the retained trade sequence is randomly reordered 10,000 times (trade-order permutation), holding the set of trade returns fixed, to characterize the distribution of maximum drawdown under alternative, equally likely historical orderings, following standard practice in systematic strategy evaluation.

4. Results

4.1 Baseline Performance

Net of transaction cost, the unfiltered base signal produced a CAGR of 12.3%, an annualized Sharpe ratio of 1.38, a Sortino ratio of 2.59, a Calmar ratio of 1.06, and a maximum drawdown of 11.6% over the sample period, with a win rate of 52.4% and a profit factor of 1.27.

4.2 Pullback-Confirmation Threshold Sweep

Table 1 reports performance across a range of confirmation thresholds θ, applied to the combined (long and short) trade population.

θ (%)	Trades Retained	% Retained	CAGR (%)	Sharpe	Max DD (%)
0.00 (baseline)	3,908	100	12.3	1.38	-11.6
0.05	3,744	96	22.3	2.37	-11.2
0.10	3,466	89	28.2	2.91	-11.0
0.15	3,163	81	31.1	3.25	-10.9
0.20	2,775	71	31.2	3.27	-10.9
0.25	2,448	63	31.9	3.33	-10.8
0.30	2,094	54	30.2	3.26	-10.8
0.40	1,541	39	24.7	3.13	-10.6
0.50	1,107	28	19.1	2.68	-10.5

Table 1. Performance across pullback-confirmation thresholds, net of transaction cost.

Performance improves monotonically from the unfiltered baseline up to θ ≈ 0.25%, where Sharpe peaks at 3.33 (a 141% relative improvement over baseline) and Sortino reaches 10.11 (baseline 2.59), before gradually declining as further tightening of the threshold reduces the retained sample to a point where fewer, higher-conviction trades no longer compensate, in aggregate compounding terms, for the reduction in trading frequency.

4.3 Robustness

The improvement is not concentrated at a single, narrow threshold. Sharpe exceeds 3.2 across the entire 0.15% to 0.35% band (Table 1 and interpolated values), a range wide enough that the result is unlikely to reflect fitting to a single, favorable

data point. Trade-order permutation (10,000 resamples) of the θ = 0.25% configuration places the realized historical maximum drawdown (10.8%) at the 87th percentile of the simulated distribution (i.e., worse than 87% of random reorderings of the same trades), indicating the historical result was not an unusually fortunate sequencing of an otherwise weaker underlying trade population.

4.4 Directional Decomposition

Applying the identical methodology separately to long and short trades shows the confirmation effect holds in both directions, though of different magnitude: long trades improve from a baseline Sharpe of 1.41 to a peak of 2.09 (θ ≈ 0.10 to 0.15%), while short trades improve from a materially weaker baseline (Sharpe 0.44) to a peak of 2.58 (θ ≈ 0.25 to 0.35%), the largest relative improvement of either leg. This is consistent with the confirmation filter disproportionately screening out lower-conviction entries, an effect that is naturally more visible in the segment (short trades, in this sample) that started with more low-conviction entries to screen out.

5. Discussion

5.1 Candidate Mechanism

We propose three, non-mutually-exclusive explanations for the observed improvement. First, a mechanical entry-price effect: by construction, every retained trade is entered at a price that has already moved in the trade’s eventual favor relative to the original signal price, which alone improves realized average return per trade, holding the signal’s directional accuracy constant. Second, an adverse-selection or confirmation effect: a signal that survives an initial adverse move before proceeding in the predicted direction may reflect stronger underlying conviction (e.g., more committed order flow, less likely to be

transient noise) than a signal that moves directly and immediately in the predicted direction, which could equally reflect a short-lived, low-conviction move. Third, a selection effect on signal quality: if the base signal's true edge is heterogeneous across trades, and lower-edge trades are more likely to move immediately without a pullback (because they lack the conviction to sustain an initial retracement test), the confirmation filter acts as an implicit, non-parametric quality filter, which is consistent with the disproportionate improvement observed on this study's weaker (short) leg.

Distinguishing empirically between these three mechanisms is beyond the scope of this paper and is a natural direction for future work, for instance by testing whether the effect persists when the entry price is held fixed at the original signal price (isolating the selection effect from the mechanical entry-price effect).

5.2 The Cost of Confirmation: Foregone Trades

At the optimal threshold identified in this sample ($\theta = 0.25\%$), 37% of signals are never converted into trades because the required pullback never materialized before the original exit. These are, by construction, trades that would have been profitable at immediate entry; the technique deliberately forgoes them in exchange for improved average quality and a better risk-adjusted profile on the trades it does take. Whether this trade-off is favorable is an empirical question specific to each strategy's trade-return distribution and cannot be assumed to generalize; a strategy whose edge derives primarily from a small number of large, immediate moves (rather than from the interaction between conviction and pullback frequency documented here) could plausibly be harmed by this technique rather than helped.

5.3 Practical Implementation and Limitations

- This is a backtested simulation using historical MAE data, not a live-execution study; actual limit-order fill rates and any additional slippage on the confirmation fill itself are not modeled and should be validated before capital deployment.
- The method assumes intrabar price continuity through the confirmation level, a reasonable approximation at hourly resolution that may not hold at lower (e.g., daily) bar frequency, where a gap could jump past the confirmation level without a valid fill opportunity, or at much higher (tick) frequency, where the underlying MAE recording granularity would need to match.
- The specific optimal threshold ($\theta \approx 0.25\%$ in this sample) is a property of this particular signal's trade-return distribution, its typical holding period, and the volatility regime of the instrument tested; it should not be treated as a universal constant, and any application to a different strategy or instrument requires re-estimating the threshold sweep documented in Section 4.2 on that strategy's own data.
- As with any single-sample historical backtest, results are subject to the general risks of regime change, structural market shifts, and, if the technique becomes widely adopted, potential erosion of the effect through crowding.

6. Conclusion

This paper isolates entry timing as a distinct, testable layer separate from both signal generation and microstructure-level execution optimization, and proposes a simple, MAE-based methodology for evaluating a specific entry-timing rule: delaying entry until a signal has been confirmed by a threshold adverse price move. Applied to a real systematic trading signal over a fifteen-year sample, the technique produced a large, cost-adjusted improvement in risk-adjusted performance

(Sharpe 1.38 to 3.33; Sortino 2.59 to 10.11) that is robust across a range of thresholds and survives Monte Carlo resequencing analysis, at the acknowledged cost of foregoing a meaningful share of trades that would have been profitable without the confirmation requirement.

The broader implication is that entry timing deserves treatment as its own research question in systematic trading, distinct from and complementary to signal design, particularly given how inexpensively it can be tested on data that most systematic trading operations already collect. We encourage other practitioners to apply the methodology in Section 3.3 to their own trade logs; the technique requires no new data infrastructure beyond what MAE tracking already provides, and, as this paper demonstrates, the result can be large enough to warrant the investigation.

References

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